Ameur Gargouri

Computer Engineer | Computer Science Teaching Candidate
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Profile

Computer Engineering graduate and applied researcher seeking a university teaching role. Combines project and research experience in programming, AI, embedded systems, computer vision, and robotics with peer-reviewed research and technical conference presentation. Interested in connecting computer science concepts with practical, project-based examples.

Relevant Teaching Areas

Programming (C, C++, Python); Linux; embedded systems; web and mobile application development.

Teaching Experience

Contractual Instructor Present
ESC Sfax, Sfax, Tunisia

- Teaching Operating Systems and Microsoft Excel to first-year students.

Computer Science Skills

Programming: C, C++, Python, Java **AI/Data:** PyTorch, scikit-learn, NumPy, pandas, OpenCV, YOLO
Embedded/Robotics: STM32, ESP32, Raspberry Pi, FreeRTOS, Embedded C, ROS2, PID, sensor fusion
Software/Web/Mobile: Git, Linux, Flutter, Node.js, MongoDB, Firebase, Laravel, Spring Boot
Methods: Continual and incremental learning, deep learning, domain adaptation

Research Experience

Human-Robot Collaboration Research 2025–2026
ATISP Laboratory, University of Sfax / ESME Research Lab Tunisia / France

- Developed a research framework for safe, personalized human-robot collaboration spanning task recognition, danger-state detection, incremental learning, and behavioral adaptation.
- Built reproducible multimodal data pipelines for HARMONIC, DASIG, and MultiPhysio-HRC, integrating physiological, inertial, video, and robot-state data.
- Proposed FLAIR, combining feature modulation, replay, Fisher regularization, and retrospective boosting; achieved forgetting of $F = 0.017$ with a 1.28 MB memory footprint.
- Trained deep 1D models on MIMU signals for danger-state detection, achieving over 98.6% macro-F1 and reducing false alarms by 87.5%.

Applied Project Experience

End-of-Studies Project – Autonomous Indoor Mobile Robot Feb–Jun 2025
ESME Research Lab, Paris, France

- Designed an autonomous mobile robot for indoor navigation and obstacle avoidance using ROS2, Raspberry Pi, FreeRTOS, sensor fusion, and YOLO.

Embedded Systems Intern – Indoor Air Quality Monitoring Jun–Aug 2024
Hydatis, Tunis, Tunisia

- Developed an STM32-based system to monitor temperature, humidity, and air pollution, with a Flutter app for real-time data visualization.

Mobile Development Intern – Olive Leaf Disease Detector Jul–Aug 2023
CRNS, Sfax, Tunisia

- Built a Flutter disease-detection app and Node.js/MongoDB backend using image processing and machine learning.

Education

PhD in Science and Technologies of Information and Communication – Year 1, in progress 2025–Present
ATISP Laboratory, University of Sfax / ESME Research Lab, Thesis: *Human-Robot Collaboration, Incremental Learning, and Behavioral Adaptation*
Computer Engineering Degree: National Engineering School of Sfax (ENIS) 2022–2025
Preparatory Classes in Mathematics and Physics: IPEIS, Sfax 2020–2022

Selected Research Output

- *Design and Development of an Autonomous Mobile Robot for Unstructured Indoor Environments*, *Machines*, 13(11), 1044 (2025), DOI: 10.3390/machines13111044.
- Presented FLAIR research at KES Conference 2026.

Additional

Certifications: NVIDIA Introduction to Deep Learning; Kaggle Intermediate Machine Learning
Languages: Arabic (native), French (B2), English (intermediate), German (elementary)